

Gaurav Rizal

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Profile

Full-stack developer and aspiring AI researcher with hands-on experience in Machine Learning, Computer Vision, LLMs, and NLP. Skilled in Python, TensorFlow, PyTorch, OpenCV, and modern web frameworks such as React. Experienced in building scalable applications and integrating ML solutions into production systems. Motivated to pursue graduate research in machine learning, computer vision, natural language processing, and multimodal systems.

Education

Kathmandu University, Bachelor of Engineering: Computer Engineering **Aug 2018 – May 2023**

- CGPA: 3.57/4.0
- Coursework: Machine Learning, Artificial Intelligence, Statistics and Probability, Data Structures & Algorithms, Digital Signal Processing, Software Engineering, Advanced Calculus, Linear Algebra, C , C++

Research Experience

Multilabel Classification of Images using MobileNetV2 **Feb 2022 – May 2022**

Active Learning Lab, Kathmandu University

github.com/physicistgaurav/Fruitstaggering

- Implemented MobileNetV2 with pre-trained ImageNet weights, achieving **95% accuracy** on the test dataset
- Converted the model to **TensorFlow Lite (TFLite)** for deployment in a Flutter mobile app, enabling object recognition via camera, photo, or URL
- Integrated with **Firestore** to provide detailed, real-time information on detected objects

Professional Experience

Full-Stack Developer, *WebNep – Urlabari, Morang* **Sept 2025 – Present**

- Currently developing a full-stack grocery management system with customer and admin platforms, including inventory, sales, payments, and credit management, tailored for a wholesale client.

Teacher, *Pashupati Awasiya Ma. Vi – Urlabari, Morang* **Feb 2025 – Aug 2025**

- Taught Computer Science and Mathematics to students of grades 9–12, fostering analytical and problem-solving skills.
- Designed engaging lesson plans, practical assignments, and assessments aligned with the national curriculum.

Software Developer, *Ramailo Tech – Gwarko, Kathmandu* **Sept 2023 – Jan 2025**

- Built LLM and speech-driven applications including a virtual assistant and an award-winning City Discussion Dashboard using GPT-4o-mini and Phi-3 Mini
- Engineered an OCR pipeline using AWS Textract, enabling automated extraction of vehicle document fields and improving data processing accuracy.
- Led AI development for the Co-Hire app using CV summarization and LinkedIn vacancy matching.

React Native Developer, *E-Signature Pvt. Ltd. – Kalopul, Kathmandu* **July 2023 - Oct. 2023**

- Contributed to Mengedegna, a bus ticketing mobile application featuring multi-ticket booking, secure payment integration, and voucher code functionality
- Responsible for complete release cycle from internal testing to Google Play Store deployment of 'Mengedegna'.

Intern, *Nepal Can Code – Tinkune, Kathmandu* **Apr 2023 – Jul 2023**

- Developed React Native frontend for BayuPayu ERP's 3 major modules (E-learning, Expense Tracker, Ticketing) including UI implementation and API integration.

Skills

AI & Machine Learning: PyTorch, TensorFlow, OpenCV, LlamaIndex, Hugging Face, OpenAI APIs, Scikit-learn, Numpy, Pandas, Seaborn, NLTK, Spacy

Programming: Python, JavaScript, TypeScript, Node.js, React.js, Next.js, Django

Cloud & Devops: AWS, Cloudflare, Firebase, Docker, CI/CD

Miscellaneous: Redis, SQL, REST APIs, Monorepo, Agile Development, TDD

Soft Skills: Adaptability, Communication, Analytical Thinking, Cross-Cultural Collaboration

Languages: English (Fluent), Nepali (Native), Hindi(Fluent)

Projects

Resume RAG github.com/Agentic-rag

- AI-powered resume analysis assistant using Gemini 2.0 Flash + LlamaIndex with multi-step reasoning for skill extraction, candidate comparison, and summarization.
- **Tools used:** LlamaIndex, Gemini, Streamlit

City Discussion Dashbaord github.com/city-discussion

- Developed a full-stack Django + React dashboard to track city-specific news and Reddit discussions. Integrated NewsAPI, PRAW, and OpenRouter for automated topic extraction, sentiment analysis, and summarization.
- **Tools used:** Python, Django, React (Vite), NewsAPI, PRAW, OpenRouter (GPT-4o-mini, Phi-3 Mini)

Seq2Seq DE-EN Translator github.com/de-en-seq2seq

- German-English translation model using Seq2Seq architecture with Bahdanau attention. Built with PyTorch, achieving meaningful translations on Multi30k dataset.
- **Tools Used:** PyTorch, spaCy , Multi30k dataset

Latte-da POS System [video-url](#)

- Developed hotel POS system with web (React) + mobile (React Native) frontend for table reservations, orders, thermal printing, sales, and credit management.
- **Tools used:** React, React Native, Node.js, Express, Prisma, TypeScript

Varenya Foundation varenyafoundation.org.np/

- Developed and deployed Varenya Foundation's complete portfolio website, managing end-to-end process from client requirements gathering to hosting, while proposing optimized solutions.
- **Tools Used:** ReactJS, Tailwind, Firebase, Vercel

Leadership Community Involvement

Leo Club of Kathmandu Himalayas Patan [2024 - 2025](#)

Acting IT Head

lckhp.org/

Kathmandu University Computer Club [2021 - 2022](#)

Dart Community Coordinator

kucc.ku.edu.np/